

CHAPTER – 08

Dictionary

What is a Dictionary?

A dictionary in Python is a collection data type used to store data in key-value pairs. It is:

- **Ordered** (as of Python 3.7+)
- Changeable (mutable)
- Does not allow duplicate keys

◆ Syntax:

```
student = {  
    'name': 'ABC',  
    'age': 20,  
    'city': 'Jaipur'  
}
```

Key Features:

- Keys must be **unique** and **immutable** (strings, numbers, tuples).
- Values can be **any data type**, including lists, tuples, and even other dictionaries.

Accessing Dictionary Items

- Using key indexing:

```
print(student['name']) # Output: ABC
```

- Using get() method:

```
print(student.get('name')) # Output: ABC
```

Modifying a Dictionary

Update Existing Value:

```
student['name'] = 'RAM'
```

Add New Key-Value Pair:

```
student['subject'] = 'English'
```

Update Multiple Pairs:

```
student.update({'age': 21, 'city': 'Delhi'})
```

Dictionary Methods:

Method	Description
keys()	Returns all keys
values()	Returns all values
items()	Returns all key-value pairs as tuples
get(key)	Access value safely
update()	Update or add key-value pairs
pop(key)	Removes item with specified key
popitem()	Removes the last inserted item
Del	Deletes item by key or entire dict
clear()	Empties the dictionary
fromkeys(keys, value)	Creates new dict from list of keys with same value

◆ Nested Dictionary

A dictionary inside another dictionary.

```
student = {  
    'name': 'X',  
    'age': 18,  
    'marks': {  
        'english': 85,  
        'math': 90,  
        'science': 75  
    }  
}
```

◆ Dictionary with Tuple & List

```
info = {  
    'roll_no': (1, 2, 3),  
    'grades': ['A', 'B', 'C']  
}
```

Practice Set (Questions Only)

1. Create a dictionary student and add keys like name, age, subject.
2. Access and print the value of the subject key using the get() method.
3. Update the age value in the dictionary.
4. Add a new key marks with value 95 to the dictionary.
5. Delete the key subject from the dictionary.
6. Use the pop() method to remove the marks key from the dictionary.
7. Create two dictionaries and merge them into a single one.
8. Given a string, count the occurrence of each character using a dictionary.
9. Write a program to check if two sets have any common elements using dictionary logic.
10. From a list of dictionaries, find the one with the highest value of a specified key (e.g., 'score').
11. Create a dictionary using fromkeys() with keys ["name", "age", "city"] and value 0.
12. Iterate through a dictionary and print all key-value pairs.
13. Demonstrate the use of popitem() and explain when it should be used.
14. Explain with code why dictionary keys must be immutable.
15. Write a program to count the number of occurrences of each element in a given list using a dictionary.